

第2讲 (2026.9.15)

①

Thm 列紧性定理

Pf $\bigcup_{n=1}^{\infty} N_{1/n} \text{ dense } X$
 $\left\{ \begin{array}{l} \text{对任意 } x \in X, \forall n, \exists x_n \in N_{1/n} \text{ s.t.} \\ d(x_n, x) < \frac{1}{n} \\ \Rightarrow x_n \rightarrow x \end{array} \right.$
 $(\because \# N_{\frac{1}{n}} < \infty)$

(M, ρ) 列紧性定理

$C(M) \stackrel{\text{def}}{=} \{ f: M \rightarrow \mathbb{R} \mid f \text{ 连续} \}$

$d(f, g) \stackrel{\text{def}}{=} \max_{x \in M} |f(x) - g(x)|$

$\Rightarrow (C(M), d)$ 完备度量空间

Def 称 $C(M)$ 中一族函数 $\mathcal{F}$ 等度连续是指:

$\forall \varepsilon > 0, \exists \delta > 0 \text{ s.t.}$

$|f(x') - f(x'')| < \varepsilon, \quad \forall x', x'' \in M \text{ with } \rho(x', x'') < \delta$

$\forall f \in \mathcal{F}$

Thm (Arzela-Ascoli)

(2)

$$\mathcal{F} \subset C(M) \text{ 列紧} \iff \mathcal{F} \text{ 作为函数族} \begin{cases} \text{一致有界} \\ \text{等度连续} \end{cases}$$

Pf: 1° " $\Rightarrow$ "

(i) 一致有界

$$\mathcal{F} \text{ 列紧} \Rightarrow \mathcal{F} \stackrel{\text{bdd}}{\subset} C(M)$$

$$\Rightarrow \mathcal{F} \text{ 作为函数族一致有界}$$

$$\sup_{f \in \mathcal{F}} d(0, f) \leq R$$

$$\iff \sup_{f \in \mathcal{F}} \max_{x \in M} |f(x)| \leq R.$$

(ii) 等度连续

$$\mathcal{F} \text{ 列紧} \Rightarrow \text{完全有界}$$

$$\Rightarrow \forall \varepsilon > 0, \exists N_{\varepsilon/3} = \{f_1, \dots, f_m\} \\ \text{s.t.}$$

$$\mathcal{F} \subset \bigcup_{k=1}^m B(f_k, \varepsilon/3)$$

$$\forall f_k, \exists \delta_k > 0 \text{ s.t.}$$

$$|f_k(x') - f_k(x'')| < \varepsilon/3,$$

$$\forall x', x'' \in M \text{ with } \rho(x', x'') < \delta$$

$$(\because f_k|_M \text{ 一致连续}(\frac{\varepsilon}{3}))$$

(3)

$$\delta \stackrel{\text{def}}{=} \min_{1 \leq k \leq m} \delta_k$$

$$\Rightarrow |f_k(x') - f_k(x'')| < \varepsilon/3$$

$$\forall x', x'' \in M \text{ with } \rho(x', x'') < \delta$$

$$\forall k \in \{1, \dots, m\}$$

$$\textcircled{1} \quad \mathcal{F} = \bigcup_{k=1}^m B(f_k, \varepsilon/3)$$

$$\Rightarrow \forall f \in \mathcal{F}, \exists k \in \{1, \dots, m\} \text{ s.t.}$$

$$d(f, f_k) < \varepsilon/3$$

$$\Rightarrow |f(x') - f(x'')|$$

$$\leq \underbrace{|f(x') - f_k(x')|}_{\leq d(f, f_k) < \varepsilon/3} + \underbrace{|f_k(x') - f_k(x'')|}_{< \varepsilon/3 \text{ if } \rho(x', x'') < \delta} + \underbrace{|f_k(x'') - f(x'')|}_{\leq d(f, f_k) < \varepsilon/3}$$

$$< \varepsilon, \quad \forall x', x'' \in M \text{ with } \rho(x', x'') < \delta$$

2° " $\Leftarrow$ "

$$\forall \varepsilon > 0, \exists \delta > 0 \text{ s.t.}$$

$$|f(x') - f(x'')| < \varepsilon/4, \quad \forall x', x'' \in M \text{ with } \rho(x', x'') < \delta$$

$$\forall f \in \mathcal{F}$$

M 列紧 $\Rightarrow \exists M$ 有穷 δ -网

(4)

$$N_\delta = \{x_1, \dots, x_n\}$$

定义映射 $T: \mathcal{F} \rightarrow \mathbb{R}^n$

$$f \mapsto (f(x_1), \dots, f(x_n))$$

$\mathcal{F}$ 一致有界

$$\Rightarrow R \stackrel{\text{def}}{=} \sup_{f \in \mathcal{F}} \max_{x \in M} |f(x)| < +\infty$$

$$\Rightarrow \left(\sum_{k=1}^n |f(x_k)|^2 \right)^{1/2} \leq \sqrt{n} R, \quad \forall f \in \mathcal{F}$$

$$\Rightarrow T(\mathcal{F}) \stackrel{\text{bdd}}{\subset} \mathbb{R}^n$$

$$\Rightarrow T(\mathcal{F}) \text{ 列紧}$$

$$\exists \tilde{N}_{\varepsilon/4} = \{Tf_1, \dots, Tf_m\} \stackrel{\text{is}}{\subset} T(\mathcal{F}) \text{ 的 } \varepsilon/4\text{-网}$$

Claim $\{f_1, \dots, f_m\} \stackrel{\text{is}}{\subset} \mathcal{F}$ 是 ε -网

$$\forall f \in \mathcal{F}, \exists k \in \{1, \dots, m\} \text{ s.t.}$$

$$d_{\mathbb{R}^n}(Tf, Tf_k) < \varepsilon/4$$

(5)

$$\forall x \in M, \exists x_j \in N_\delta \text{ s.t.}$$

$$\rho(x, x_j) < \delta$$

 $\Rightarrow$

$$|f(x) - f_k(x)|$$

$$\leq \underbrace{|f(x) - f(x_j)|}_{< \varepsilon/4} + \underbrace{|f(x_j) - f_k(x_j)|}_{\leq d_{\mathbb{R}^n}(Tf, Tf_k)} + \underbrace{|f_k(x_j) - f_k(x)|}_{< \varepsilon/4}$$

$$(\because \rho(x, x_j) < \delta$$

$$< \varepsilon/4$$

由等度连续($\frac{\varepsilon}{4}$)

$$< \frac{3}{4} \varepsilon$$

$$\Rightarrow d(f, f_k) < \varepsilon$$

Q: L^p 中 $\mathcal{F}$ 列 "紧" 性 判定 据 ?

Thm (Riesz - Fréchet - Kolmogorov)

$$\text{设 } 1 \leq p < \infty$$

$\mathcal{F} \subset L^p$ 列 "紧" $\Leftrightarrow$ 且 仅 当

$$(i) \sup_{f \in \mathcal{F}} \|f\|_p < \infty \quad (\text{i.e. } \mathcal{F} \stackrel{\text{bdd}}{\subset} L^p)$$

(ii) 等度 L^p 收敛.

$\forall \varepsilon > 0, \exists R > 0$ s.t.

$$\int_{|x| > R} |f(x)|^p dx < \varepsilon^p, \quad \forall f \in \mathcal{F}$$

(iii) 等度 L^p 范数连续.

$\forall \varepsilon > 0, \exists \delta > 0$ s.t.

$$\|\tau_h f - f\|_p < \varepsilon, \quad \forall h \in \mathbb{R} \text{ with } |h| < \delta$$
$$\forall f \in \mathcal{F}$$

where

$$(\tau_h f)(x) = f(x+h)$$

H. Brezis << F.A., Sobolev spaces and PDE >>

Thm 4.26. Cor. 4.27

Remark: L^p 中更常用弱收敛性.

例: 设 $1 \leq p < \infty$,

$$A \subset \ell^p \text{ 弱收敛} \Leftrightarrow \begin{cases} \text{(i)} A \stackrel{\text{bdd}}{\subset} \ell^p \\ \text{(ii)} \forall \varepsilon > 0, \exists N \text{ s.t.} \\ \sum_{k=N+1}^{\infty} |x_k|^p < \varepsilon^p \\ \forall x \in A. \end{cases}$$

(7)

Pf: 1° " $\Rightarrow$ " (HW)2° " $\Leftarrow$ "只需证明 A 完全有界 $\forall \epsilon > 0$, 令

$$\epsilon = (2^{p+1} + 1)^{-\frac{1}{p}}$$

 $\Rightarrow \exists N$ s.t.

$$\sum_{k=N+1}^{\infty} |x_k|^p < \epsilon^p, \quad \forall x \in A$$

$$\vec{x} \mapsto T: \ell^p \rightarrow \mathbb{R}^N$$

$$(x_1, x_2, \dots) \mapsto (x_1, x_2, \dots, x_N)$$

$$A \not\subset_{\ell^p} \Rightarrow \exists M \text{ s.t.}$$

$$\sum_{k=1}^{\infty} |x_k|^p \leq M^p, \quad \forall x \in A$$

$$\Rightarrow |x_k| \leq M, \quad \forall k, \quad \forall x \in A$$

$$\Rightarrow \left(\sum_{k=1}^N |x_k|^2 \right)^{1/2} \leq \sqrt{N} M$$

$$\forall x \in A$$

$$\Rightarrow T(A) \stackrel{\text{bdd}}{\subset} \mathbb{R}^N$$

$$\Rightarrow \exists T(A) \approx \frac{\varepsilon}{N} - (\hat{x})$$

(8)

$$\{Tx^{(1)}, \dots, Tx^{(m)}\}$$

Claim $\{x^{(1)}, \dots, x^{(m)}\} \stackrel{?}{\approx} A \approx \gamma \vec{x}$

$$\forall x \in A, \exists j \in \{1, \dots, m\} \text{ s.t.}$$

$$d_{\mathbb{R}^n}(Tx, Tx^{(j)}) < \frac{\varepsilon}{N}$$

$$= \left(\sum_{k=1}^N |x_k - x_k^{(j)}|^2 \right)^{1/2}$$

$$\Rightarrow |x_k - x_k^{(j)}| < \frac{\varepsilon}{N}, \quad k=1, 2, \dots, N$$

$$\Rightarrow \|x - x^{(j)}\|_p^p$$

$$= \underbrace{\sum_{k=1}^N |x_k - x_k^{(j)}|^p}_{< N \cdot \left(\frac{\varepsilon}{N}\right)^p} + \underbrace{\sum_{k=N+1}^{\infty} |x_k - x_k^{(j)}|^p}_{\leq 2^p \sum_{k=N+1}^{\infty} (|x_k|^p + |x_k^{(j)}|^p)}$$

$$< N \cdot \left(\frac{\varepsilon}{N}\right)^p$$

$$\leq 2^p \sum_{k=N+1}^{\infty} (|x_k|^p + |x_k^{(j)}|^p)$$

$$< 2^p \cdot 2\varepsilon^p$$

$$< (2^{p+1} + 1) \varepsilon^p$$

$$\Rightarrow \|x - x^{(j)}\|_p < \gamma$$

HW: 1° Hilbert cube

$$Q = \{x \in \ell^2 : |x_k| < 2^{-k}, \forall k\}$$

$$\frac{1}{2} \ell^2 \not\subset Q \quad \left| \sum_{k=1}^{\infty} \frac{1}{2^k} \right|$$

2° Ex. 1.3.5, 1.3.6.